Worksheet — 1.2 Software & Software Development

OCR A Level Computer Science (H446) — Component 01, Section 1.2 (1.2.1 Operating systems · 1.2.2 Applications generation · 1.2.3 Software development · 1.2.4 Types of programming language)

Name: ___ *Date:* ___

Time: ~60 min. Check against the answer key at the end.

Section A — Skills practice

1. Key-term match

Write the letter of the correct definition next to each term.

Term	Answer		Definition
1. Virtual memory	____	A	Excessive paging between RAM and disk that slows the system almost to a halt.
2. Interrupt	____	B	Software emulating a complete computer, or running platform-independent intermediate code.
3. Disk thrashing	____	C	Using secondary storage as if it were RAM to run more/larger programs than RAM allows.
4. ISR	____	D	Dividing memory into fixed-size blocks (pages mapped to frames).
5. Paging	____	E	A signal (hardware or software) that a device/process needs the processor's attention.
6. Virtual machine	____	F	The OS code (handler) run to deal with one particular interrupt.

2. The four stages of compilation (in order)

Fill in the blanks. Write the stages **in the correct order**.

Stage 1 — _____ analysis: source code is broken into __; whitespace and comments are removed; a __ table is built.

Stage 2 — __ *analysis (also called* ___): tokens are checked against the language's __; a __ tree is built; __ _____ errors are reported here.

Stage 3 — Code ______: the parse tree and symbol table are used to produce __ / machine code.

Stage 4 — Code ______: the code is refined to run __ or use less __, while producing the same result.

3. Translators — complete the comparison table

	Compiler	Interpreter
When does translation happen?	________	________
Output produced?	________	________
Program execution speed	________	________
Source code needed to run?	________	________
How are errors reported?	________	________
Best used for…	________	________

Q. In one sentence, what does an **assembler** translate, and at roughly what ratio?

4. Match the methodology to the scenario

Methodologies: **Waterfall · Agile · Extreme Programming (XP) · Spiral · RAD**

Write the best-fit methodology next to each scenario.

a) A bank is building a large, expensive air-traffic adjacent system where the cost of failure is huge; management wants formal **risk analysis** at every iteration. → ________

b) A small project for a client whose requirements are **fixed, clear and unlikely to change**, and who wants full documentation at each stage. → ________

c) A start-up needs a working version **fast** and is happy to refine **prototypes** with users; requirements are vague. → ________

d) Safety-critical control software where **code quality is paramount**, so the team uses **pair programming** and test-driven development with a customer on-site. → ________

e) An app whose requirements are **expected to change a lot**; the client wants to see and feed back on **working increments** regularly. → ________

5. Addressing modes

A program executes the instruction `LDA 20`. Memory contents are:

Address	Contents
20	35
35	99
99	12

State the value that is loaded into the accumulator under each addressing mode.

a) **Immediate** addressing: value loaded = ______

b) **Direct** addressing: value loaded = ______

c) **Indirect** addressing: value loaded = _______

d) In one line, explain what **indexed** addressing uses to find the operand, and what it is typically used for.

6. Scheduling & interrupts — short answer

a) Name the scheduling algorithm that gives each process a fixed **time slice (quantum)** in turn, returning unfinished jobs to the back of the queue. State whether it is pre-emptive.

b) **Shortest Job First** minimises average waiting time. Give **one** disadvantage.

c) Define **pre-emptive** scheduling.

d) Put the interrupt-handling steps in order (write 1–6):

Step	Order (1–6)
Load the address of the correct ISR	_____
Pop the registers off the stack and resume the original task	_____
CPU checks the interrupt register at the end of the F-D-E cycle	_____
Run the ISR	_____
Push the registers (e.g. PC, ACC) onto the stack	_____
Finish the current instruction (if the interrupt is higher priority)	_____

e) Why must the registers be saved on a **stack** (LIFO) rather than in a single store?

7. OOP — terms labelling/matching

Match each OOP term to its meaning.

Terms: **Class · Object · Method · Attribute · Inheritance · Encapsulation · Polymorphism**

Meaning	Term
A specific instance of a class.	__________
A template/blueprint defining attributes and methods.	__________
A variable belonging to a class/object (its data/state).	__________
A subroutine belonging to a class (its behaviour).	__________
Bundling data and methods together and hiding the data (private attributes accessed via methods).	__________
A subclass acquiring the attributes/methods of a superclass, and able to add or override them.	__________
The same method name behaving differently depending on the object.	__________

8. Challenge box

Challenge. A team is distributing finished commercial software to customers who must **not** see the source code, and the program must **run quickly** on the user's machine without needing anything else installed.

(i) Should they use a **compiler** or an **interpreter**? Justify with **two** reasons drawn from the table in Q3.

(ii) The OS uses **virtual memory** to let several large programs run at once. Explain why virtual memory does **not** actually increase the amount of RAM, and name the problem caused by **excessive** swapping.

9. Interrupt handling with a nested interrupt — order the sequence

A word processor is running when the printer raises an interrupt (out of paper). While the printer's ISR is running, a **higher-priority** power-failure interrupt occurs. Number the steps **1–7** in the order they happen. Examiners report that candidates often forget the priority check and the reverse-order restore — watch for both.

____ The registers are popped off the stack in reverse order and the word processor resumes exactly where it left off.

____ At the end of the current FDE cycle, the CPU checks the interrupt register and finds a flag set.

____ The contents of the registers (PC, ACC, etc.) are pushed onto the stack.

____ The PC is loaded with the address of the printer's ISR (from the interrupt vector), and the ISR begins to run.

____ The printer interrupt's priority is compared with the current task's; it is higher, so it will be serviced.

____ The power-failure interrupt occurs: the printer ISR's register values are pushed onto the stack, and the higher-priority ISR runs to completion first.

____ The printer ISR's registers are popped off the stack and the printer ISR finishes running.

10. Which methodology — and why?

For each scenario, choose the best-fit methodology **(Waterfall · Agile · Extreme Programming (XP) · Spiral · RAD)** and — unlike activity 4 — add a **one-sentence justification that quotes evidence from the scenario**. A bare methodology name with no scenario-linked reason scores nothing.

Scenario	Methodology	Justification (quote the scenario)
a) A games studio must show a publisher a playable demo within weeks, then evolve it sprint by sprint using tester feedback.		
b) Tax-return software must implement fully specified, legally fixed government rules, with formal documentation and sign-off at each stage.		
c) A large defence project must evaluate the risks of each phase, building a prototype before more budget is committed to the next loop.		
d) A two-developer team writing firmware for a heart-rate monitor: quality is critical, a customer representative is available daily, and they release small, constantly-tested increments.		
e) An internal admin tool is needed urgently; users will refine throwaway prototypes with the developers, and a polished user interface matters more than efficient code.		

Section B — Exam practice (30 marks)

Answer all questions. Marks and assessment objectives are shown in brackets.

Q1 State **two** functions carried out by an operating system. **[2]** (AO1)

Q2 Describe what happens during the **lexical analysis** stage of compilation. **[2]** (AO1)

Q3 An A-level student is learning to program in Python. She writes short programs, makes small changes and re-runs her code many times each lesson. Explain why a language that is **interpreted** is well suited to the student's situation. **[3]** (AO2)

Q4 A charity has asked a developer to build a fundraising website. The trustees are not sure exactly what features they want, expect their ideas to change as they see the site develop, and would like to try out working versions regularly. Explain why an **agile** approach is more appropriate than the **waterfall** lifecycle for this project. **[4]** (AO2)

Q5 A processor executes the instruction `LDA 40`. The index register currently holds **3**. Memory contents are:

Address	Contents
40	55
43	21
55	88

Complete the table to show the value loaded into the accumulator under each addressing mode. **[4]** (AO2)

Addressing mode	Value loaded into ACC
Immediate	
Direct	
Indirect	
Indexed	

Q6 State what is meant by an **interrupt**, and give **one** example of an event that would generate one. **[2]** (AO1)

Q7 Describe **two** differences between **paging** and **segmentation**. **[4]** (AO1)

Q8* A company is developing the control software for a fleet of autonomous delivery drones. The software is safety-critical: a fault could cause a drone to crash in a public place. However, the client also admits that some requirements — such as how the drones will be tracked by customers — are still evolving and will change during the project.

Discuss the suitability of the **waterfall lifecycle** and of **agile methodologies** for this project, and recommend how the company should proceed. **[9]** (AO3)

The quality of your extended response will be assessed in this question.